

ABSTRACT

A **Basic Banking System** is a software-based solution designed to automate and simplify the core operations of a bank. It enables efficient management of customer accounts, transactions, fund transfers, and balance inquiries while reducing the need for manual paperwork. This system is widely used in banks, credit unions, and other financial institutions to ensure smooth and secure handling of day-to-day banking activities.

The system operates by securely storing customer data, account details, and transaction history in a centralized database. Users can perform essential banking functions such as opening or closing accounts, depositing or withdrawing money, and checking account balances through a user-friendly interface—either at bank branches, ATMs, or through online and mobile banking apps. This not only speeds up banking operations but also reduces the chances of human errors.

One of the key features of the Basic Banking System is its ability to handle digital transactions efficiently. Customers can transfer funds between accounts, pay utility bills, or receive salary credits with ease. Every transaction is recorded with time stamps and unique identifiers to ensure transparency and traceability. The system also includes authentication methods like PINs, OTPs (one-time passwords), and biometric verification to ensure secure access and prevent fraud.

In addition, the software generates real-time reports and summaries that help bank employees and managers monitor cash flow, customer activity, and overall financial performance. These analytics support better decision-making in areas like loan approvals, interest rate adjustments, and fraud detection. The automation of routine tasks like passbook updates, interest calculation, and statement generation leads to reduced workloads for staff and improved service delivery for customers.

By providing 24/7 access to banking services and enhancing operational efficiency, a Basic Banking System plays a critical role in modernizing financial services. As the demand for digital banking continues to grow, such systems are essential for promoting financial inclusion, improving customer experience, and building trust in the banking sector.

CHAPTER -1

INTRODUCTION

1.1-Problem Statement

Traditional banking systems, especially in small-scale or rural setups, often rely on manual record-keeping or basic spreadsheet tools. These outdated methods come with several drawbacks and operational challenges, such as:

- Difficulty in tracking customer transactions, deposits, and withdrawals
- Manual interest calculation and balance updates leading to errors and delays
- Limited or no facility to generate financial reports or summaries
- Higher chances of fraud, data loss, or unauthorized access due to lack of security

There is a need for an efficient, accurate, and easily maintainable system that can handle bank operations in an organized and automated manner.

1.2 Description

The Basic Bank System is a backend-focused project built using **MySQL** to demonstrate the application of DBMS principles in a real-world context. It includes multiple tables (*Account*, *Books*, *Transactions*, *Fines*) that are linked through foreign key relationships to enforce rules and minimize redundancy.

Features of the system include:

- Creating and managing customer account records
- Performing deposits, withdrawals, and fund transfers
- Automatic balance updates and fine calculations
- Viewing account statements and transaction history

CHAPTER -2

OBJECTIVE

The objective of this project is to design and implement a **Basic Banking System** using a **relational database approach**. The system aims to automate and streamline essential banking operations such as:

- Maintaining records of customer accounts
- Handling deposits, withdrawals, and fund transfers
- Tracking transaction history and balances
- Calculating and managing fines or penalties
- Ensuring data accuracy, consistency, and secure retrieval

This project is also intended to help students understand and apply core **DBMS concepts**, including **table creation, normalization, SQL queries, relationships, and constraints**.

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CHAPTER -3

REQUIREMENTS

3.1-Software Requirements

Component	Specification
Database System	MySQL Server (v8.0+) or MySQL Workbench
Backend (Optional)	Python with MySQL Connector
Operating System	Windows 10+, Linux, or macOS
IDE/Text Editor	VS Code, Eclipse, or MySQL Workbench
Browser (Optional GUI)	Google Chrome / Firefox (for future web UI)

3.2-Hardware Requirements

- **Processor:** Intel Core i3 or equivalent
- **RAM:** Minimum of 2GB
- **Disk Space:** Minimum of 500 MB
- **Display:** Standard display for IDE visibility
- **Input Device:** Keyboard for user input in the console

CHAPTER -5

IMPLEMENTATION & RESULTS

OUTPUT

5.1- SQL TABLES:

```

*****
<><><><><><><>[ MAIN MENU ]<><><><><><>
*****
1.OPEN NEW ACCOUNT
2.DEPOSIT AMOUNT
3.WITHDRAW AMOUNT
4.BALANCE ENQUIRY
5.ACCOUNT DETAIL
6.CLOSE ACCOUNT
*****
*****

Enter Your Choice >>> 1
*****OPEN NEW ACCOUNT*****
Enter Account No.:- 23456
Enter Name :- Jhon
Enter Date of Birth :- 10/01/1979
Enter Address :- Bangalore
Enter Mobile No.:- 9865739803
Enter Opening Balance :- 10000
Congratulaions Account Open Successfully

```

```

*****
<><><><><><><>[ MAIN MENU ]<><><><><><>
*****
1.OPEN NEW ACCOUNT
2.DEPOSIT AMOUNT
3.WITHDRAW AMOUNT
4.BALANCE ENQUIRY
5.ACCOUNT DETAIL
6.CLOSE ACCOUNT
*****
*****

Enter Your Choice >>> 1
*****OPEN NEW ACCOUNT*****
Enter Account No.:- 34567
Enter Name :- Alice
Enter Date of Birth :- 17/05/1980
Enter Address :- Mangalore
Enter Mobile No.:- 8375893670
Enter Opening Balance :- 30000
Congratulaions Account Open Successfully

```

```

MySQL localhost:3306 ssl banking_system SQL > select * from account;
+-----+-----+-----+-----+-----+-----+
| Acc_no | Name  | Dob       | Address  | Mob_no   | Balance |
+-----+-----+-----+-----+-----+-----+
| 23456  | Jhon  | 10/01/1979 | Bangalore | 9865739803 | 10000   |
| 34567  | Alice | 17/05/1980 | Mangalore | 8375893670 | 30000   |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.0011 sec)

MySQL localhost:3306 ssl banking_system SQL > select * from amount;
+-----+-----+-----+
| Acc_no | Name  | TotalBalance |
+-----+-----+-----+
| 23456  | Jhon  | 10000         |
| 34567  | Alice | 30000         |
+-----+-----+-----+
2 rows in set (0.0009 sec)

```

```

*****
<><><><><><><><>[ MAIN MENU ]<><><><><><>
*****

1.OPEN NEW ACCOUNT
2.DEPOSIT AMOUNT
3.WITHDRAW AMOUNT
4.BALANCE ENQUIRY
5.ACCOUNT DETAIL
6.CLOSE ACCOUNT
*****
*****

Enter Your Choice >>> 2
*****DEPOSIT AMOUNT*****
Enter Amount :- 500
Enter Account No :- 23456

```

```

2 rows in set (0.0009 sec)
MySQL localhost:3306 ssl banking_system SQL > select * from amount;
+-----+-----+-----+
| Acc_no | Name  | TotalBalance |
+-----+-----+-----+
| 23456  | Jhon  | 10500        |
| 34567  | Alice | 30000        |
+-----+-----+-----+
2 rows in set (0.0011 sec)

```

```

*****
<><><><><><><><>[ MAIN MENU ]<><><><><><>
*****

1.OPEN NEW ACCOUNT
2.DEPOSIT AMOUNT
3.WITHDRAW AMOUNT
4.BALANCE ENQUIRY
5.ACCOUNT DETAIL
6.CLOSE ACCOUNT
*****
*****

Enter Your Choice >>> 3
*****WITHDRAW AMOUNT*****
Enter Withdraw Amount :- 500
Enter Account No :- 34567

```

```

2 rows in set (0.0011 sec)
MySQL localhost:3306 ssl banking_system SQL > select * from amount;
+-----+-----+-----+
| Acc_no | Name  | TotalBalance |
+-----+-----+-----+
| 23456  | Jhon  | 10500        |
| 34567  | Alice | 29500        |
+-----+-----+-----+

```

CHAPTER -6

CONCLUSION

The **Basic Banking System** provides a robust and scalable solution for automating daily operations of a banking service. Using a relational database model in **MySQL with MySQL Connector**, the project effectively manages data related to **customer accounts, transactions, and fines**. It improves accuracy, reduces manual workload, and ensures the integrity of banking records.

This project not only demonstrates key concepts of **Database Management Systems** like **normalization, constraints, and relationships** but also prepares students for real-world application development. It can be easily extended with a **graphical user interface or web integration** using tools such as **MySQL Connector**

With clear structure, practical utility, and extendable design, this project is an excellent foundation for students learning **DBMS**.

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CHAPTER -8

REFERENCES

1. *Fundamentals of Database Systems*, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.
2. *Database Management Systems*, Ramakrishnan and Gehrke, 3rd Edition, 2014, McGraw Hill
3. ChatGPT